

Engineering Mechanics 2021

Polytechnic Study Hub - Downloadable Handbook Notes

Course Code 2021, Course Title Engineering Mechanics, Revision 2021

Semester 2, Credits 3, Category Engineering Science, Periods per week 3 [L2 T1 P0], Total periods 45

Programmes: Architecture, Automobile, Civil allied, Mechanical allied, Wood and Paper Technology, Environmental Engineering.

Official source: supplied syllabus PDF. Official model question paper pattern is not specified in the supplied source; model paper here is fresh practice.

Official Overview

Objective: impart knowledge and skills to predict effects of force and motion.

Prerequisites: Integration and trigonometry from Mathematics I and II; scalar/vector quantities and triangular/parallelogram law of forces from Applied Physics I.

CO1 Identify force systems using basics of mechanics, 10 h, Understanding. CO2 Apply static equilibrium for unknown forces in structural elements, 12 h, Applying.

CO3 Solve rigid-body problems using distributed areas and masses, 11 h, Applying. CO4 Determine structural behaviour of materials under loading, 10 h, Applying.

CO-PO mapping: CO1 PO1=2; CO2 PO1=3 and PO2=2; CO3 PO1=3; CO4 PO1=3. 3 strong, 2 moderate, 1 weak.

Module 1 - Force Systems

Mechanics studies bodies under force and motion. Statics means equilibrium; dynamics means motion. Use particle, rigid body and flexible body idealisations correctly.

Scalar has magnitude only; vector has magnitude and direction. Force resolution: $F_x = F \cos \theta$, $F_y = F \sin \theta$, $R = \sqrt{F_x^2 + F_y^2}$, $\tan \theta = F_y / F_x$.

Principle of transmissibility: force can move along its line of action without changing external effect on a rigid body. Moment $M = F d$, unit N m.

Varignon theorem: moment of resultant equals sum of moments of components. Equilibrium needs $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$.

FBD is compulsory: show body isolated, applied loads, reactions, dimensions and chosen axes. Lami theorem applies only for three coplanar concurrent non-parallel forces.

Module 2 - Beams, Trusses, Friction

Supports: simple/roller gives one reaction, hinge gives horizontal and vertical reactions, fixed support gives reactions plus fixing moment.

Loads: point load, inclined point load, UDL and couple. UDL equivalent $W=wL$ acts at centroid of loaded length, not automatically at full beam centre.

Beam reactions: convert UDL, mark distances, use $\sum M=0$ and $\sum F_y=0$. Simply supported beam central load W gives $R_A=R_B=W/2$; cantilever end load W gives fixed moment WL .

Truss method of joints: find reactions, choose joint with two unknowns, assume tension, apply $\sum F_x=0$ and $\sum F_y=0$; negative answer means compression.

Friction: limiting friction $F=\mu R$, $\tan \phi=\mu$, angle of repose equals angle of friction at impending motion.

Module 3 - Centroid and MI

Centroid is geometric centre of area; centre of gravity is point through which weight acts. For uniform lamina they coincide.

Composite centroid steps: split into simple shapes, choose axes, prepare A, x, y, Ax, Ay table, use $\bar{x} = \frac{\sum Ax}{\sum A}$ and $\bar{y} = \frac{\sum Ay}{\sum A}$. Holes are negative areas.

Standard centroid: rectangle centre, triangle $h/3$ from base, semicircle $4r/[3\pi]$ from diameter, quarter circle $4r/[3\pi]$ from both axes.

Moment of inertia measures distribution of area resisting bending. Radius of gyration $k = \sqrt{I/A}$. Parallel axis theorem $I = I_G + Ad^2$. Perpendicular axis theorem $I_z = I_x + I_y$.

For composite MI, use own centroidal MI plus Ad^2 for each part; subtract holes. Polar MI of solid circular section $J = \pi d^4/32$.

Module 4 - Stress and Strain

Stress $\sigma = P/A$, unit N/mm^2 or MPa. Strain is deformation/original length, no unit.

Hooke law: stress proportional to strain within elastic limit.

Young modulus $E = \text{stress/strain}$. Axial elongation $\Delta L = PL/[AE]$. Convert load, length and area units before substitution.

Types of stress: direct tensile/compressive, shear, bending and torsional. Know nature and direction of each stress.

Mild steel stress-strain curve: proportional limit, elastic limit, yield point, ultimate stress and breaking point. Tor steel uses proof stress because yield plateau is not clear.

Elastic constants: E , G , K and Poisson ratio ν . Relations: $E = 2G[1 + \nu]$, $E = 3K[1 - 2\nu]$, $E = 9KG/[3K + G]$. Properties: ductility, malleability, brittleness, toughness, stiffness, hardness.

Formula Bank and Solved Examples

Resultant: $R_x = \sum F_x$, $R_y = \sum F_y$, $R = \sqrt{R_x^2 + R_y^2}$, $\theta = \tan^{-1}[R_y/R_x]$. Moment: $M = F \cdot d$.

Equilibrium: $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$.

UDL: $W = wL$ at $L/2$ from start of UDL. Friction: $F = \mu R$, $\tan \phi = \mu$. Centroid:

$\bar{x} = \sum Ax / \sum A$, $\bar{y} = \sum Ay / \sum A$. MI: $I = I_G + Ad^2$, $k = \sqrt{I/A}$.

Stress formulas: $\sigma = P/A$, $\text{strain} = e/L$, $E = \sigma / \text{strain}$, $\delta = PL/AE$. Units: $1 \text{ N/mm}^2 = 1 \text{ MPa}$.

Solved 1: 100 N at 30 deg gives $F_x = 86.6 \text{ N}$ and $F_y = 50 \text{ N}$. Solved 2: 5 m simply supported beam with 10 kN central load gives $R_A = R_B = 5 \text{ kN}$.

Solved 3: 20 mm dia rod under 30 kN has $A = 314.16 \text{ mm}^2$, $\text{stress} = 30000/314.16 = 95.5 \text{ N/mm}^2$.

Solved 4: $\mu = 0.35$ gives $\phi = \tan^{-1}[0.35] = 19.3 \text{ deg}$.

Practice, Model Paper and References

Expected questions: define statics, vector, moment, centroid, MI, stress, strain; explain Varignon theorem, Lami theorem, FBD, support types, friction laws, stress-strain curve.

Numericals: resultant of forces, Lami theorem, beam reactions for point load and UDL, truss by joints, coefficient of friction, composite centroid, MI by parallel axis theorem, axial stress and elongation.

Fresh model paper: Time 3 hours, Max marks 100. Part A 10x2 all modules. Part B answer any 8 of 5 marks. Part C answer any 4 of 10 marks. Pattern is practice only because official pattern was not specified in source.

Answer guide: draw diagram/FBD first, write formula, substitute with units, calculate, state final answer and member nature if truss. Avoid wrong UDL centroid, inconsistent moment sign and unit mixing.

References specified: Khurmi Applied Mechanics, Bansal Engineering Mechanics, Ramamrutham Engineering Mechanics, Bedi Strength of Materials, Khurmi SOM, Punmia SOM, Meriam and Kraige Statics, Rattan SOM, NPTEL 122104015 and 112103109.